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CS 621 Data Analytics
Peer-Reviewed Project

Data Analytics Project Plan: Academic Performance Data

Introduction

Student academic performance is an important factor in education. Educators, parents, and students share common concerns about what is being taught and how that learning translates into application. In this Data Analytics Project Plan, we will present our approach to analyzing student academic performance data, including data source information, data ingestion, data engineering, analytic computation, visualizations, and expected deliverables.

Data Source

The dataset for this project is the "Students Academic Performance Dataset" found on the Kaggle website (https://www.kaggle.com/datasets/sadiajavedd/students-academic-performance-dataset). The data is in a clean CSV format and contains information on 1000 students, including eight variables of interest (See Figure 1 for a small sample of the dataset):

1. Gender
2. Race/Ethnicity
3. Parental Level of Education
4. Lunch (free/reduced or standard)
5. Test Preparation Course (completed or none)
6. Math Score
7. Reading Score
8. Writing Score

The test score variables are the results of three standardized tests in math, reading, and writing that each of the 1000 students completed. Analyzing this dataset using the given variables will provide insight into relationships among certain categories (gender, socioeconomic status, test preparation) and the resulting outcomes for academic performance on these standardized tests.

Time needed: 1 hour

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

Figure 1: Sample of Dataset

Data Ingestion

During the data ingestion stage, the data collected from Kaggle will be stored in a local CSV file. Once the data is loaded into a Python program (utilizing Jupyter Lab), the following data structures will be used to store the raw data:

- Lists
- Dictionaries
- Pandas Series (See Figure 2)
- Pandas DataFrame

Time needed: 5 hours

```
gender  race/ethnicity
female  group A           36
        group B          104
        group C          180
        group D          129
        group E           69
male    group A           53
        group B           86
        group C          139
        group D          133
        group E           71
dtype: int64
```

Figure 2: Pandas Series

Data Engineering

During the data engineering stage, the following work will be done:

- Identifying and removing NA or null values
- Identifying and removing duplicated student entries
- Ensuring consistency within categorical variable nomenclature

Time needed: 3 hours

Analytic Computation

During the analytic computation stage, we will first conduct basic analytics of the data to include:

- Number of students
- Number of race groups
- Counts for levels of parental education
- Counts for types of lunch received (standard and free/reduced)
- Counts for test preparation course (completed and none)
- Basic summary statistics (average, percentiles, etc.) for each test score

Additionally, during the analytic computation stage, we will conduct advanced analytics to include:

- Comparisons between gender and each test score
- Comparisons between socioeconomic status based on the lunch variable and each test score
- Comparisons between parental education level and test scores
- Comparison of the test preparation course and above-average scores on each of the tests
- Comparisons among test scores

Time needed: 18 hours

Visualization

During the visualization stage, we will rely upon the results from the analytic computation stage to present graphs of the findings. Some graphs that we will provide include:

- Boxplots to display test score comparisons (See Figure 3)
- Bar Charts to display the comparisons for above-average test scores (See Figure 4)
- Scatter Plots to display comparisons among the test scores

Time needed: 22 hours

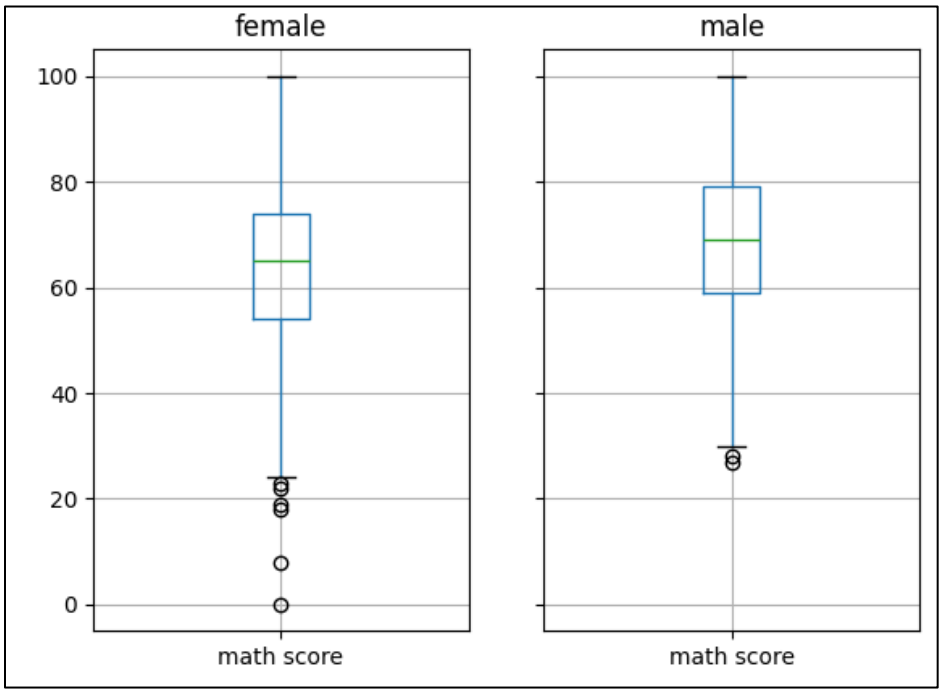


Figure 3: Box Plots Comparing Math Scores Across Gender

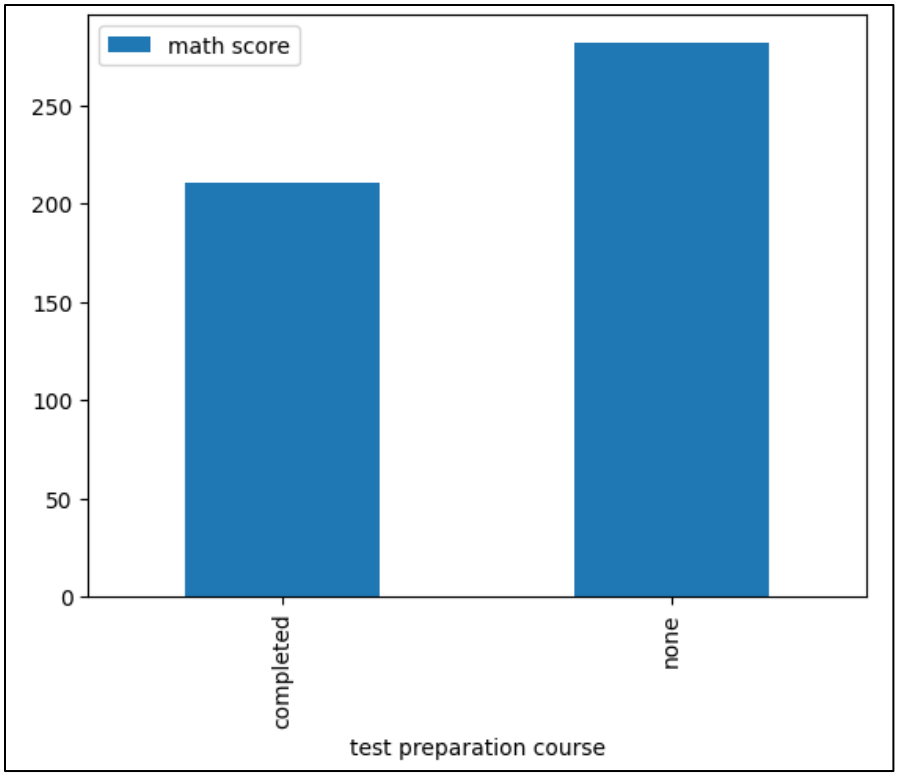


Figure 4: Bar Chart Comparing Test Preparation and Math Score

Deliverables

Once the analysis is complete, we will deliver the following items:

- Cleaned dataset (1000 rows)
- A written report (6 – 10 pages)
- A PowerPoint slide deck (15 slides)
- A 15-minute recorded presentation (MP4)
- A Python program for interactive demonstration (~500 lines of code with documentation presented in a Jupyter Notebook format)

Time needed: 15 hours